

CHRIS SCHMIDT

Baltimore, Maryland | p.christopher.schmidt@gmail.com | 410-409-3105
linkedin.com/in/chris-schmidt-94526a10 | github.com/PCSchmidt | pcschmidt.github.io

PROFESSIONAL SUMMARY

Machine Learning Engineer with five years of data, software, and ML engineering at HII Mission Technologies, supporting DoD data modernization on AWS GovCloud. Active security clearance. Master's degree in Applied Mathematics; pursuing an MS in AI Engineering at Johns Hopkins University. Builds AI systems where the consequential decision stays in deterministic code and model output is independently checked or measured, shown through original projects with live demos and published evaluations.

TECHNICAL SKILLS

Programming Languages: Python, SQL, R, SAS (professional); C++, Rust, Go, TypeScript/JavaScript, Shell scripting, Astro (personal projects)

Cloud Platforms: AWS GovCloud (Glue, S3, Athena, Lambda, RDS, DynamoDB, Kinesis, EFS, SQS, Elasticsearch), Azure, GCP, Databricks, Advana

Data & AI Engineering: ETL/ELT Pipelines, Data Ingestion & EDA, PowerBI, Qlik Sense, RAG architectures, RAGAS evaluation, vector search (pgvector/Supabase), evals-driven development, MLOps

Machine Learning: PyTorch, TensorFlow, Scikit-learn, BERT, NLP, OCR, Deep Learning, computer vision (YOLO, ByteTrack, TensorRT), Kalman filtering, DBSCAN, IsolationForest, geometric deep learning (exploratory)

AI/GenAI & Agentic Tooling: LLM application development, LangChain, LangGraph, Claude API, Claude Code, prompt engineering, agentic harness & multi-agent system design, Cursor, Codex

DevOps & Infrastructure: Docker, Kubernetes, Git/GitHub Actions, GitLab CI/CD, Terraform (IaC, exploratory), Fly.io, Vercel, Railway

Frameworks & Collaboration: FastAPI, Next.js, React/React Native (Expo), Supabase, Linux/Bash, Jira, Confluence, SharePoint

KEY GOVERNMENT PROJECTS

F-35 Data Modernization (DoD/JPO): Leading analytics work to achieve audit-worthy data architectures for the Joint Program Office, developing Python-based analysis tools and executive dashboards.

FORTIS AI Program (DoD): Developed data processing pipelines and integrated BERT-based NLP models to enhance AI capabilities for defense applications.

Advana Platform Analytics (DoD): Conducted enterprise-scale data quality analysis using Databricks and Qlik Sense, collaborating with MILDEP and IA stakeholders.

Center for Program Integrity (CMS): Performed big data analytics for fraud detection and program integrity using SAS and Python.

PERSONAL AI ENGINEERING PROJECTS

Self-directed systems built on one idea: keep consequential decisions in deterministic code, and independently check or measure model output before it is relied on. Live demos, evaluations, and write-ups at pcschmidt.github.io.

Grey-Resolve: Degraded-Media Entity Resolution, Evaluated Honestly

- Face detection, quality gating, embedding, and FAISS retrieval over degraded media, evaluated with per-condition ROC/FMR/FNMR sweeps; the quality gate lowers FNMR, and the central context-fusion hypothesis was tested at scale and reported as not supported; 241 deterministic tests, live demo

SiteWatch AI: Computer Vision Safety Monitoring, Verified by an RLM Agent

- Built a two-plane safety monitor: YOLO11s + ByteTrack rules fire on every frame; a Prime Agent RLM verifies each firing by running Python over the raw tracks, and a deterministic gate recomputes its numbers before anything is stored
- Agent eval on 30 labeled incidents: 86.7% first pass at ~\$0.002 per incident; the gate caught a real agent misclassification on the live demo; 191 tests; Terraform validated for AWS, GCP, and Azure

Meridian: Agent Harness for AI-Assisted Development

- Harness that blocks an AI coding agent's next action when a gate fails, with an independent evaluator subagent to prevent self-grading (self-score 5.5/10 vs independent 2.5/10); 240 assertions across 18 test suites

Hard Power Intelligence: Defense/AI/Energy Intelligence Platform

- Multi-desk platform (FastAPI, Next.js, Supabase/pgvector) turning USAspending, SEC EDGAR, and NRC data into source-attributed briefings behind a fabrication-checking eval gate
- Entity resolver at 1.0 precision / 1.0 recall; passed an OWASP Top 10 security audit with zero Critical/High findings

CERTIFICATIONS

AWS Solutions Architect Associate | DeepLearning.AI Data Engineering Professional Certificate | IBM Generative AI Engineering Certificate | IBM AI Engineering Professional Certificate | Deep Learning Specialization Certificate | ML Ops | Machine Learning Operations Specialization, Duke University

SECURITY CLEARANCE

Active Security Clearance (Details available upon request)

PROFESSIONAL EXPERIENCE

MISSION TECHNOLOGIES, A DIVISION OF HII, Arlington, Virginia (Remote)

Machine Learning Engineer

October 2024 – Present

- Lead analytics for F-35 data modernization initiative (JPO), developing Python-based tools and PowerBI dashboards
- Perform enterprise-scale data architecture analysis to facilitate DoD audit-readiness transformations
- Develop Python scripts for data analysis, generating actionable recommendations for stakeholders
- Leverage Amazon GovCloud stack for secure, compliant data processing and storage

Software Engineer

May 2022 – October 2024

- Led data analysis and quality initiatives for DSCA/DoD modernization, collaborating with MILDEP and IA SMEs to align strategic goals
- Conducted exploratory data analysis (EDA) and built visualizations using Databricks and Qlik Sense within the Advana platform
- Developed data pipelines for the FORTIS AI program using Python and SQL, including API development, enhancing operational efficiency
- Created Python tools for complex EDA incorporating OCR, language translation, geospatial analysis, and graph-based approaches (Cypher, SPARQL, RDF)

Data Engineer, Associate

July 2021 – May 2022

- Developed serverless ETL microservices using AWS Lambda, RDS, DynamoDB, Kinesis, S3, EFS, SQS, and Elasticsearch, processing and ingesting up to 1 million text documents per hour
- Integrated BERT-based NLP models for the FORTIS AI project, enhancing language processing capabilities
- Conducted cost optimization analysis, reducing operational expenses
- Authored technical documentation and maintained security compliance requirements

PRIOR PROFESSIONAL BACKGROUND

Business Owner, Mobtown Restoration Group

8 years

- Owned and operated a construction/restoration firm, growing annual revenue to \$1–2 million and leading a team of up to 10 employees at peak

Senior Sales Associate, Institutional Fixed Income Sales, Wachovia Securities

6 years | Series 7/63

- Managed institutional client relationships and executed fixed income sales within a regulated financial services environment

Technical Writing Consultant, Medical Textbook & Journal Publishers

4 years

- Provided technical and medical/scientific writing services on a consulting basis, supporting publisher overflow workload

Mortgage Loan Originator

2 years

- Originated mortgage loans, performing credit analysis and underwriting support for borrowers

Additional sales and client-facing roles: geotechnical engineering product sales; financial advising at Morgan Stanley; small business banking; commercial telecom sales at MCI

EDUCATION

Johns Hopkins Whiting School of Engineering

Master of Science, AI Engineering (Part-time, Expected May 2029)

Towson University

Master of Science, Applied Mathematics (Concentration: Applied Statistics)

Bachelor of Science, Applied Mathematics

Bachelor of Science, Political Science and History

LEADERSHIP

20 years as collegiate rugby head coach, including 10 years at Towson University and 10 years at Loyola University Maryland, focused on team development, strategy, and mentorship.